

Deepfake Detection

Seeing Is No Longer Believing

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CSE 200

Our Team



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Wait a second...

We never Won the nobel prize

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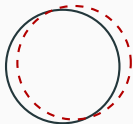
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The Threat and The Solution

What's a Deepfake?



A fake video, photo, or audio clip made by AI — made to look and sound like the real thing.

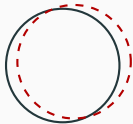
What's Detection?



Teaching AI to spot the tiny mistakes humans miss — and call out the fake.

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Can You Spot the Fake?

Real or Fake? Two of these are **AI-generated**.



Image 1



Image 2



Image 3



Image 4

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Image 1



Image 2



Image 3

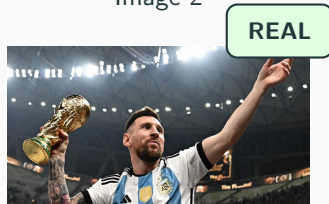


Image 4

How Are They Generated?

Source Identity

(Voice / Face)

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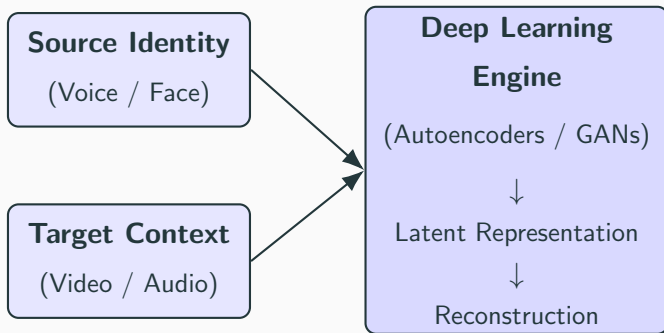
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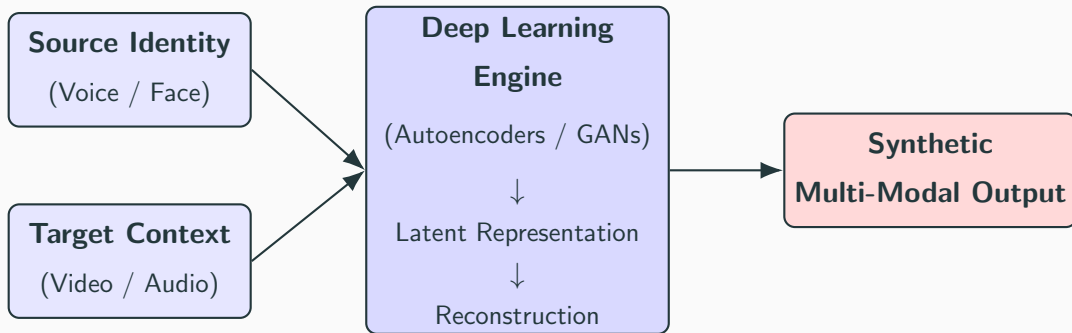
Target Context

(Video / Audio)

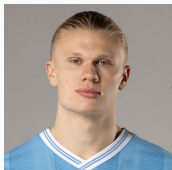
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How Are They Generated?



Example: From Inputs to Output



Source Image



Target Image



Outcome (Deepfake)

The Crisis of Trust: A Weaponized Tool

- **Geopolitical Misinfo** — fabricated speeches or events used to sway public opinion.
- **Corporate Fraud** — fake voices or video used to authorize scams and wire transfers.
- **Identity Abuse** — impersonation used for harassment or non-consensual content.
- **Erosion of Evidence** — makes real footage deniable, undermining trust in evidence itself.

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can it also be trained to catch the forgery?**



Up Next: Deepfake Detection Algorithms

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Detection Techniques

The Two Fronts of Detection

Passive Defense



Examine media *after*
creation for hidden flaws

Reactive

Proactive Defense



Embed safeguards *into* the
generator at creation time

Preventative

It's a cat-and-mouse game — every new detector eventually meets a generator built to beat it.

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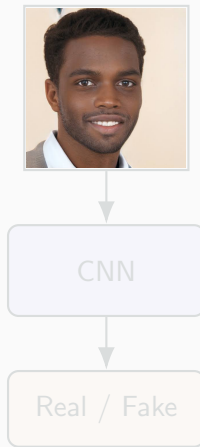
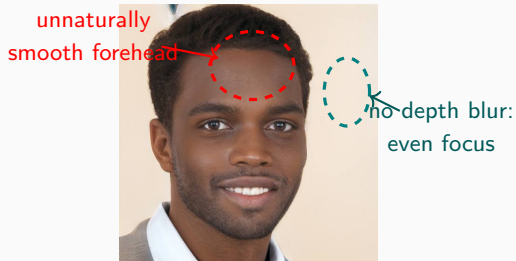


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Passive Detection: Spatial Artifacts



A single frame is enough: waxy, over-smoothed skin and an unnatural lack of depth-of-field are learned by a standard CNN.

Passive Detection: Spatial Artifacts

unnaturally
smooth forehead

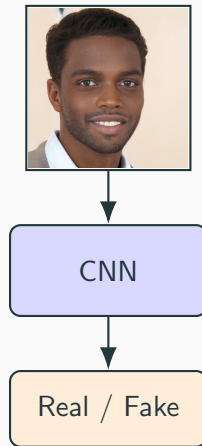
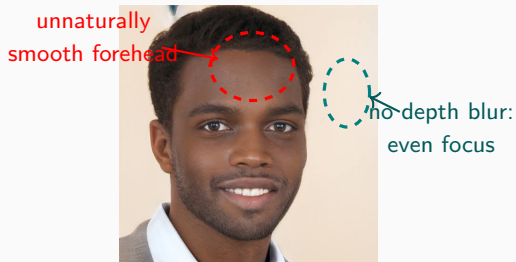
no depth

!! RED FLAGS !!



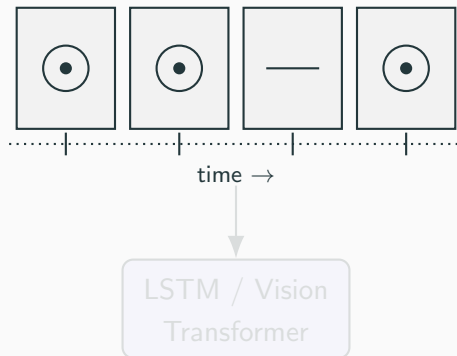
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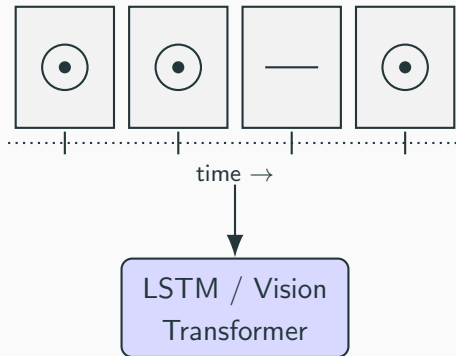
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Passive Detection: Temporal Artifacts



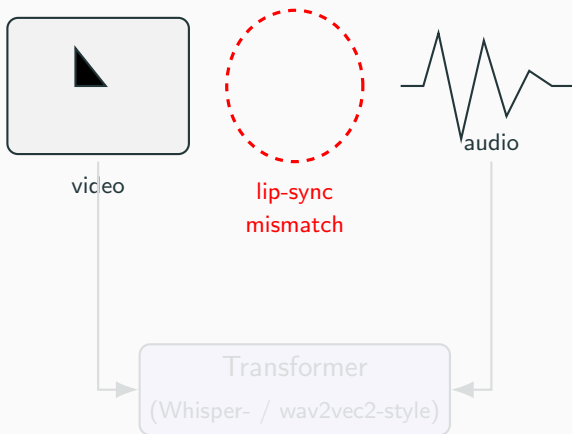
Fakes struggle to stay consistent *across* frames: unnatural blink rates, jerky motion, missing heartbeat-driven skin-color shifts.

Passive Detection: Temporal Artifacts



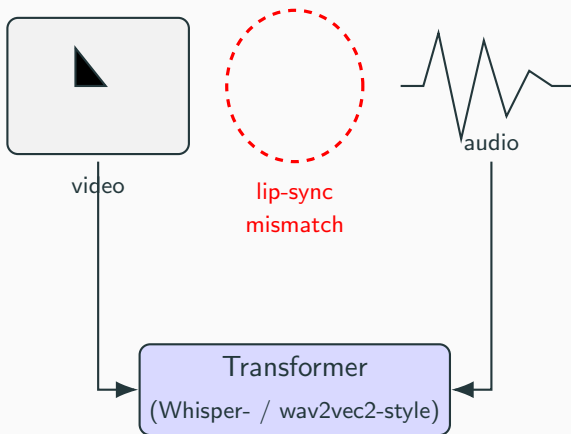
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The Multimodal Frontier: Audio Meets Video



Voice can be faked too. Cross-checking acoustic cues against mouth movement — e.g. does a “B” sound match a closed mouth? — exposes synthetic audio.

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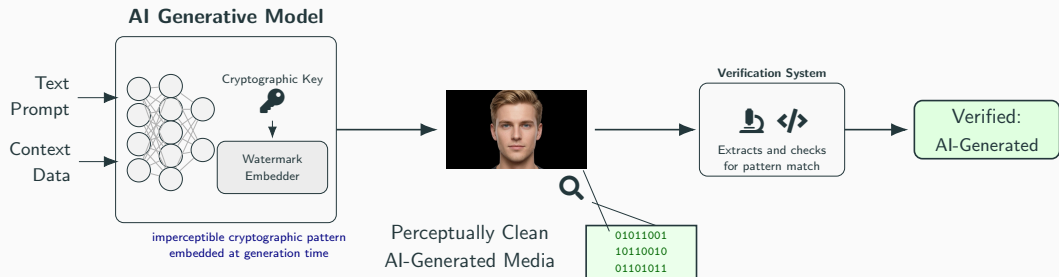
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Passive defense is a cat and mouse game



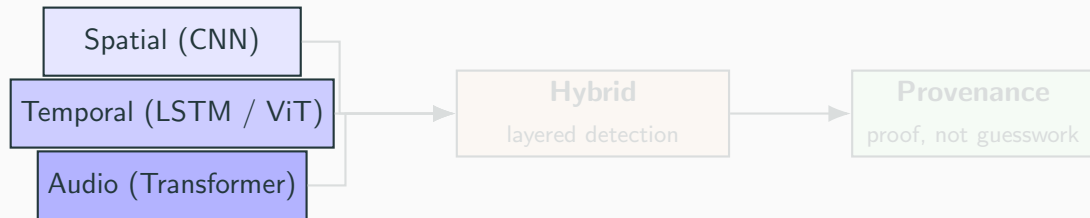
As generating models get better at creating realistic fakes, detection models must continuously be upgraded to keep pace.

Proactive Defense: Invisible Watermarking



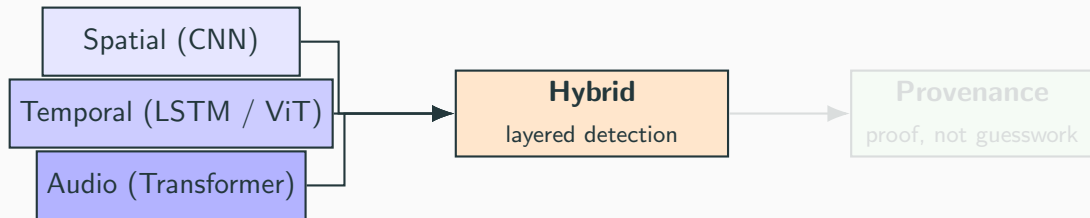
Survives compression, cropping, and screenshots — so platforms can confirm origin without any loss in quality.

Summary: No Silver Bullet



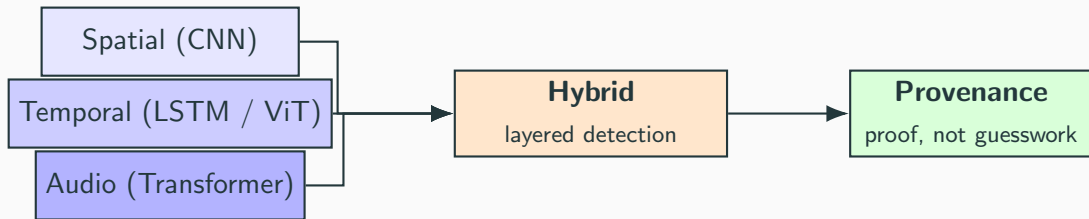
The field is shifting from playing detective — hunting for AI mistakes — to demanding cryptographic proof of authenticity.

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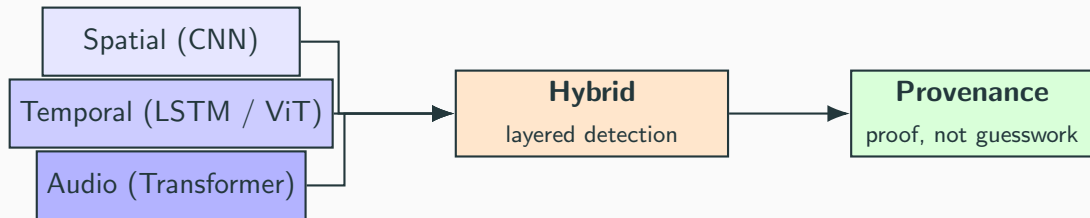
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We can now spot a fake in a single frame.

**But can our tools survive the internet, the courtroom, the
real world?**



Up Next: From Detection to Defense

Detection Works in the Lab — But Does It Survive Real Life?

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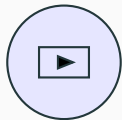
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From Detection to Defense

Where Detection Matters



Social Platforms

Upload scanning

Content labeling



FinTech & Identity

Fraud prevention

Liveness check



Courts & Forensics

Evidence auditing

Forensic verification



Govt. & Elections

Statement verification

Media integrity

Different environments *demand* different guarantees.

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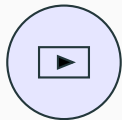


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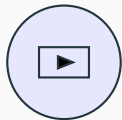


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The Future Is Not “One Detector”



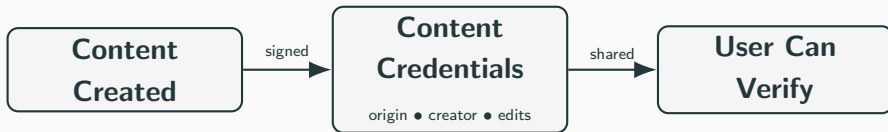
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The Future of Authenticity: Proving Where Content Came From

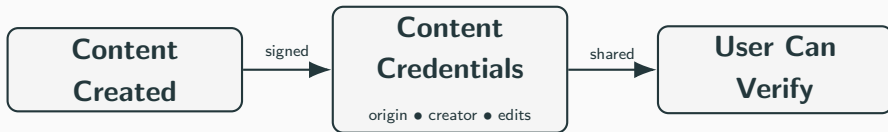


Detection asks: *"Does this look fake?"*

Provenance asks: *"Can we prove where it came from?"*

C2PA / Content Credentials — an emerging industry standard.

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The Road Ahead



Multimodal

image, video, audio



Continuous

ongoing verification



Standardized

provenance by default



Human-in-the-loop

AI flags, people decide

The goal isn't a perfect detector.

It's a world where deception is harder to spread — and easier to verify.

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Seeing is no longer believing.
Verification is.

Thank You